

GenAI. Lecture 5 - Energy models. MCMC

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LAMBDA • HSE

Rejection Sampling



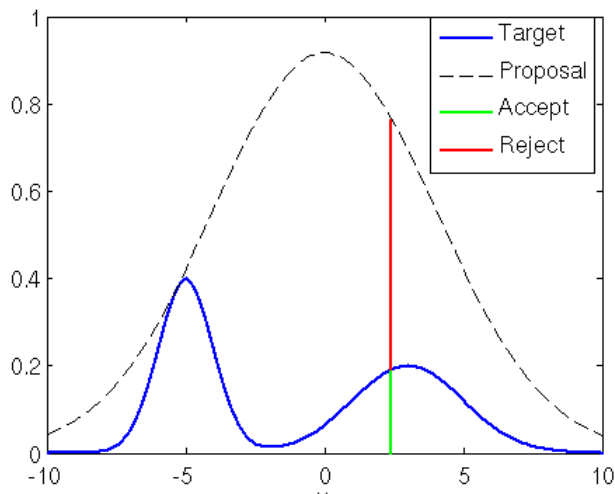
Rejection Sampling

- ▶ Rejection sampling is a basic Monte Carlo method to generate samples from a target distribution $P(x)$ using a proposal distribution $q(x)$.
- ▶ It requires a constant M such that $P(x) \leq Mq(x)$ for all x .

Rejection Sampling Algorithm

1. Sample x' from the proposal distribution $q(x)$.
2. Sample $u \sim \text{Uniform}(0, 1)$.
3. Accept x' if $u \leq \frac{P(x')}{Mq(x')}$, else reject and repeat.
4. The accepted samples follow the target distribution $P(x)$.

Rejection Sampling



Gibbs Sampling



Gibbs Sampling

- ▶ Gibbs sampling is a Markov Chain Monte Carlo (MCMC) algorithm for obtaining a sequence of observations approximated from a specified multivariate probability distribution.
- ▶ It samples each variable sequentially from its conditional distribution given the current values of the other variables.

Gibbs Sampling Algorithm

1. Initialize all variables $x_1, x_2, \dots, x_n$.
2. For each iteration:
 - Sample x_1 from $P(x_1|x_2, \dots, x_n)$
 - Sample x_2 from $P(x_2|x_1, x_3, \dots, x_n)$
 - $\vdots$
 - Sample x_n from $P(x_n|x_1, \dots, x_{n-1})$
3. Repeat until convergence.

Metropolis-Hastings Scheme



Detailed Balance Equation

- ▶ The **detailed balance equation** ensures that the probability flow between any two states is equal in both directions:

$$P(x)q(x, x') = P(x')q(x', x)$$

where $q(x, x')$ is the transition probability from state x to x' .

- ▶ This condition guarantees that the target distribution $P(x)$ is the stationary distribution of the Markov chain.
- ▶ Detailed balance is a sufficient condition for the convergence of MCMC algorithms.

Question: How to ensure the detailed balance equation for any P and q ?

Detailed Balance and Metropolis-Hastings

- Let

$$q'(x, x') = \alpha q(x', x)$$

where $\alpha = \min \left(1, \frac{P(x')q(x|x')}{P(x)q(x'|x)} \right)$.

- Ensures that the Markov chain has the desired stationary distribution $P(x)$.
- In Metropolis-Hastings, detailed balance is maintained by the acceptance probability:

$$\alpha = \min \left(1, \frac{P(x')q(x|x')}{P(x)q(x'|x)} \right)$$

Metropolis-Hastings specifics

Benefits:

- ▶ All you need is to know $P(x)$ **up to constant**
- ▶ Works with **any** proposal distribution q

Limitations:

- ▶ the sampling performance directly depends from the $q(x, x')$ choice

Example. Energy models

- ▶ Energy-based models define a probability distribution through an energy function $E(x)$:

$$P(x) = \frac{e^{-E(x)}}{Z}$$

- ▶ Z is the partition function, often intractable.

MCMC Methods



MCMC Methods

- ▶ **Langevin Sampling:** Uses gradient information to guide the sampling process.
- ▶ **Hamiltonian Sampling:** Incorporates momentum and simulates physical dynamics to explore the state space efficiently.

Langevin Sampling

- ▶ Combines random noise with gradient-based updates.
- ▶ Update rule:

$$x_{t+1} = x_t + \frac{\epsilon^2}{2} \nabla \log P(x_t) + \epsilon \cdot N(0, I)$$

Do not forget to perform Metropolis-Hastings acceptance scheme on each step!

Hamiltonian Dynamics

- ▶ The Hamiltonian $H(x, p)$ represents the total energy of the system:

$$H(x, p) = U(x) + K(p)$$

- ▶ $U(x)$ is the potential energy related to the target distribution: $U(x) = -\log P(x)$.
- ▶ $K(p)$ is the kinetic energy, often chosen as $K(p) = \frac{1}{2}p^T M^{-1}p$, where M is the mass matrix.

Hamilton's Equations

- ▶ The evolution of the system is governed by Hamilton's equations:

$$\frac{dx}{dt} = \frac{\partial H}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial x}$$

- ▶ These equations describe how position x and momentum p change over time.
- ▶ Simulating these dynamics allows the sampler to propose new states efficiently.

Leapfrog Integration



Numerical Integration: The Leapfrog Method

- ▶ Exact solutions to Hamilton's equations are often intractable.
- ▶ The leapfrog method provides a numerical approximation:
 1. $p_{t+\frac{1}{2}} = p_t - \frac{\epsilon}{2} \nabla U(x_t)$
 2. $x_{t+1} = x_t + \epsilon M^{-1} p_{t+\frac{1}{2}}$
 3. $p_{t+1} = p_{t+\frac{1}{2}} - \frac{\epsilon}{2} \nabla U(x_{t+1})$
- ▶ The leapfrog method preserves the volume and reversibility of the system.

HMC Algorithm

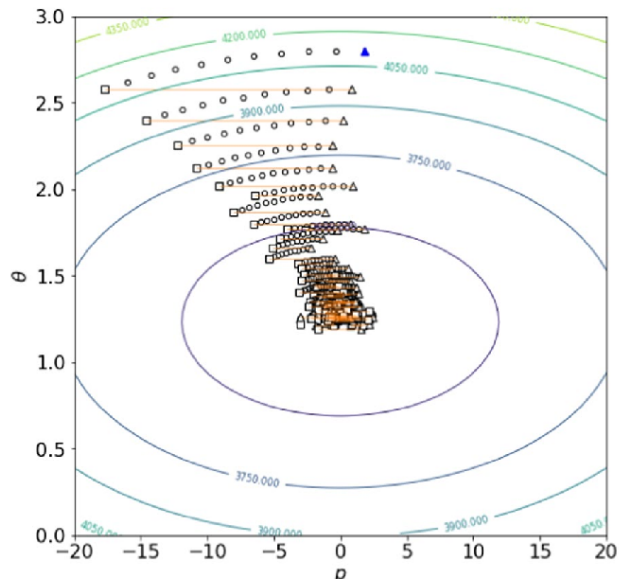


Hamiltonian Monte Carlo Algorithm

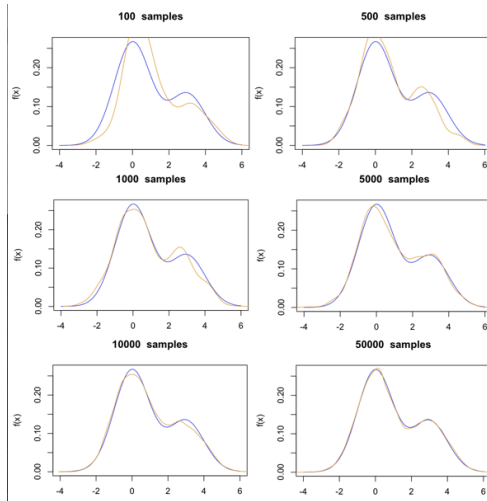
1. Initialize at position x_0 and momentum $p \sim \mathcal{N}(0, M)$.
2. For each iteration:
 - Simulate Hamiltonian dynamics using the leapfrog method.
 - Compute acceptance probability:

$$\alpha = \min(1, \exp(H(x_t, p_t) - H(x', p')))$$

- Accept or reject the new state based on α .



Illustration



Conclusion

- ▶ Gibbs sampling and Metropolis-Hastings are foundational MCMC methods.
- ▶ Langevin and Hamiltonian methods improve sampling efficiency.
- ▶ Energy models offer a flexible framework for probabilistic modeling.
- ▶ Detailed balance ensures MCMC methods converge to the correct stationary distribution.